

PAPER 1 / QUESTION 87

2224-6701 / original question 2

87. An object is launched upwards at an initial velocity $+u$ at an angle of θ to the horizontal. Air resistance is negligible.

At a later time the object has a vertical displacement of zero.

What are the horizontal component of the velocity and the vertical component of the velocity at this later time?

	Horizontal component of the velocity	Vertical component of the velocity
A.	$u \cos \theta$	$u \sin \theta$
B.	$u \cos \theta$	$-u \sin \theta$
C.	$u \sin \theta$	$u \cos \theta$
D.	$u \sin \theta$	$-u \cos \theta$